

JomExam

ESAT Physics — Sample Paper

Six questions in the style of ESAT Physics, including a circuit diagram, a velocity–time graph and a paired-options question. Allow 9 minutes — the real paper's 90-second pace. No calculator. Take $g = 10 \text{ N kg}^{-1}$. One option is correct in each question, and there is no negative marking.

Worked solutions with the trap named follow the questions. The full experience — 27 questions, 40 minutes, and a report naming the skill behind every wrong answer — is free at jomexam.com.

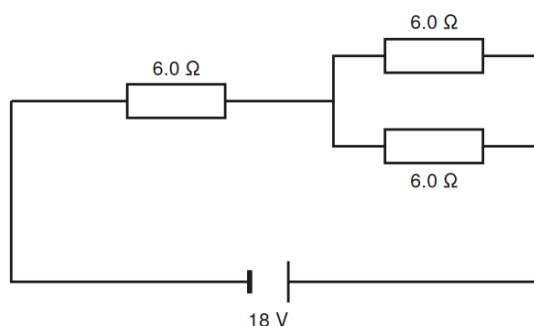
Question 1. A stone is thrown vertically upwards at 20 m s^{-1} and later returns to the thrower's hand. Taking $g = 10 \text{ N kg}^{-1}$ and ignoring air resistance, for how long is it in the air?

- A 2 s
- B 3 s
- C 4 s
- D 5 s
- E 40 s

Question 2. A cube of side 0.50 m and mass 250 kg rests on the ground. Taking $g = 10 \text{ N kg}^{-1}$, what pressure does it exert?

- A 500 Pa
- B 1 000 Pa
- C 2 500 Pa
- D 5 000 Pa
- E 10 000 Pa

Question 3. Three identical 6.0Ω resistors are connected as shown, across an 18 V supply of negligible internal resistance. What current flows from the supply?



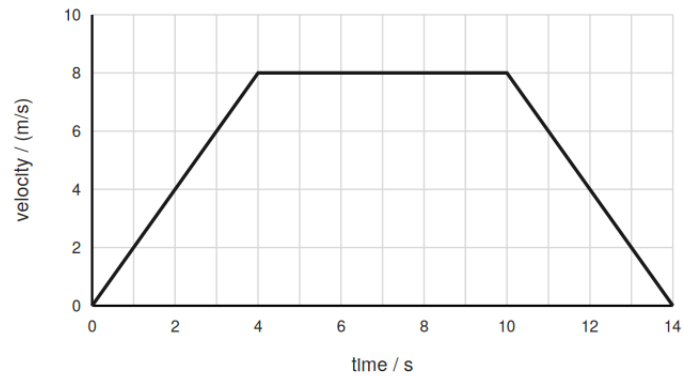
- A 1.0 A
- B 2.0 A
- C 3.0 A
- D 6.0 A
- E 9.0 A

Question 4. A pendulum bob is released from rest at a point 0.45 m above the lowest point of its swing. Taking $g = 10 \text{ N kg}^{-1}$ and ignoring air resistance, what is its speed at the lowest point?

- A 3.0 m s^{-1}
- B 4.5 m s^{-1}

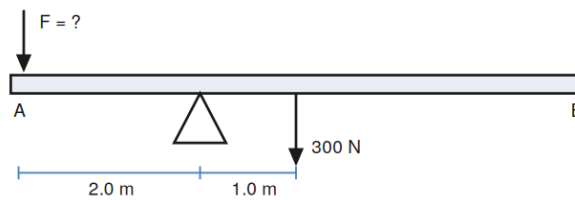
- C 9.0 m s^{-1}
- D 2.1 m s^{-1}
- E 30 m s^{-1}

Question 5. The velocity–time graph shows a cyclist's journey. What total distance does the cyclist travel in the 14 seconds shown?



- A 48 m
- B 64 m
- C 80 m
- D 96 m
- E 112 m

Question 6. A uniform beam of length 6.0 m and weight 300 N is pivoted 2.0 m from end A, as shown. A downward force F applied at end A holds the beam horizontal. Which row gives the force F , and the upward force exerted by the pivot on the beam?



	Force F at A	Force from the pivot
A	100 N	400 N
B	150 N	150 N
C	150 N	300 N
D	300 N	600 N
E	450 N	150 N
F	150 N	450 N
G	300 N	300 N
H	100 N	300 N

Worked solutions

Question 1 — answer C.

1. Time to the top: the speed falls by 10 m s^{-1} each second, so 20 m s^{-1} takes 2 s.
2. The flight is symmetric: 2 s up and 2 s down.

The trap: The trap is 2 s — the journey to the top reported as the whole flight. Symmetry is free marks: whatever goes up takes exactly as long to come down.

Question 2 — answer E.

1. Weight = $250 \times 10 = 2\,500 \text{ N}$.
2. Contact area = $0.50 \times 0.50 = 0.25 \text{ m}^2$.
3. $P = F/A = 2\,500 / 0.25 = 10\,000 \text{ Pa}$.

The trap: Two traps share the question: dividing the MASS by the area (1 000), and treating the side as the area (5 000). Pressure wants newtons over square metres, and neither arrives without being computed.

Question 3 — answer B.

1. The parallel pair: two equal resistors halve, giving $3.0 \, \Omega$.
2. In series with $6.0 \, \Omega$: total $9.0 \, \Omega$.
3. $I = V/R = 18/9.0 = 2.0 \text{ A}$.

The trap: The trap is 1.0 A, from adding all three in series ($18 \, \Omega$). A parallel pair is always LESS resistive than either branch — if your total went above $12 \, \Omega$, the mistake announced itself early.

Question 4 — answer A.

1. Energy conservation: $\frac{1}{2}mv^2 = mgh$, so $v = \sqrt{2gh}$.
2. $v = \sqrt{2 \times 10 \times 0.45} = \sqrt{9} = 3.0 \text{ m s}^{-1}$.

The trap: The trap is 9.0 — computing $2gh$ perfectly and forgetting the square root. The mass cancels before the arithmetic starts; noticing that is what buys the time back.

Question 5 — answer C.

1. Distance is the area under the graph. First 4 s: a triangle, $\frac{1}{2} \times 4 \times 8 = 16 \text{ m}$.
2. 4 s to 10 s: a rectangle, $6 \times 8 = 48 \text{ m}$.
3. 10 s to 14 s: a triangle, $\frac{1}{2} \times 4 \times 8 = 16 \text{ m}$. Total $16 + 48 + 16 = 80 \text{ m}$.

The trap: Every wrong option is an area mishandled: 48 counts only the steady phase, 64 forgets the final triangle, 112 turns both triangles into rectangles, and 96 pays full speed for 12 of the 14 seconds. Area under the line, piece by piece — nothing else survives.

Question 6 — answer F.

1. The weight acts at the beam's centre, 3.0 m from A — which is 1.0 m from the PIVOT, on the far side. Its moment: $300 \times 1.0 = 300 \text{ N m}$.
2. F acts 2.0 m from the pivot: $F = 300/2.0 = 150 \text{ N}$.
3. The pivot supports everything pushing down: $300 + 150 = 450 \text{ N}$.

The trap: Two traps, one per column. The arm for the weight is 1.0 m from the pivot, not 3.0 m from A — every arm is measured from the pivot and nowhere else. And the pivot force must carry the beam AND your applied force: rows pairing 150 N with 300 N collect everyone who balanced the moments and then forgot what they had just added to the beam.

Ready for the real pace? Ten questions in fifteen minutes, free: the JomExam mini test, at jomexam.com. © JomExam.
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